Math 3
10-2R Solve Equations Algebraically: Review

Name:_____
Date: _____ Per: ___

For problems 1 - 8, solve for x.

1) $3x - 4 = 2x + 8 - 5x$

$3x-4=-3x+8$
$+3x \quad +3x$

$6x-4=8$
$+4 \quad +4$

$\frac{6x}{6} = \frac{12}{6}$

$x=2$

2) $2 - 15x = 5(3x + 2)$

$2-15x=15x+10$
$-15x \quad -15x$

$2-30x=10$
$-2 \quad -2$

$\frac{-30x}{-30} = \frac{8}{-30}$

$x = \frac{-8}{30} \rightarrow \frac{-4}{15}$

3) $6(x - 4) = (\frac{2}{3}x - 8 + \frac{1}{6}x)6$

$6x-24=4x-48+x$

$6x-24=5x-48$
$-5x \quad -5x$

$x-24=-48$
$+24 \quad +24$

$x=-24$

4) $8x = 4(2x + 1)$

$8x=8x+4$
$-8x \quad -8x$

$0=4$

NS

5) $10(6.2x + 5 - 1.4x) = (4.8x + 6)10$

$62x+50-14x=48x+60$

$48x+50=48x+60$
$-48x \quad -48x$

$50=60$

NS

6) $10(5.4x + 14.6 - 10.1x) = (12.8x - 3.5x - 0.6)10$

$54x+146-101x=128x-35x-6$

$-47x+146=93x-6$
$-93x \quad -93x$

$-140x+146=-6$
$-146 \quad -146$

$\frac{-140x}{-140} = \frac{-152}{-140} \rightarrow \frac{38}{35}$

7) $12\left[14 - \frac{1}{4}(x - 10)\right] = \left[\frac{5}{6}(x + 15)\right]12$

$168-3(x-10)=10(x+15)$

$168-3x+30=10x+150$

$198-3x=10x+150$
$-10x \quad -10x$

$198-13x=150$
$-198 \quad -198$

$\frac{-13x}{-13} = \frac{-48}{-13}$

$x=\frac{48}{13}$

8) $2\left[2(3x + 2)\right] = \left[\frac{1}{2}(12x + 8)\right]2$

$4(3x+2)=1(12x+8)$

$12x+8=12x+8$
$-12x \quad -12x$

$8=8$

IS

9) The membership fee for joining a camping group is $45. A campsite charges $35 per night for members of this camping group and nonmembers $40 per night. After how many nights would a nonmember and a member pay the same amount?

X = # NIGHTS

MEMBER = NONMEMBER

[FEE]+[COST PER NIGHT][#NIGHTS] = [NON COST PER NIGHT][#NIGHTS]

$45 + 35x = 40x$

$-35x \quad -35x$

$\frac{45}{5} = \frac{5x}{5}$

$x = 9$ NIGHTS

10) In the 2020 NFC Championship Game, the 49ers scored 1 less than ⅔ the total points scored in the game. The Packers scored 4 more than ⅔ the total points scored in the game.

SF POINTS LEFT 2/3

Write and solve an equation to determine the total points scored in the game.

X = TOTAL POINTS

49ERS + PACKERS = TOTAL

$\left(\frac{2}{3}x - 1\right) + \left(\frac{4}{3}x - 56\right) = x$

$2x - 57 = x$

$-x \quad -x$

$x - 57 = 0$

$+57 \quad +57$

$x = 57$ PTS

11) Who won the game in item 10? What was the final score?

49ERS

$\frac{2}{3}(57) - 1$

$38 - 1$

37 PTS

PACKERS

$\frac{4}{3}(57) - 56$

$76 - 56$

20 PTS

4 9 7 3

For problems 12 - 15, simplify the expression.

12) $(x^9)^2 \cdot x^{14}$

$(x^{9 \cdot 2}) \cdot x^{14}$

$x^{18} \cdot x^{14}$

x^{32}

13) $\frac{(2x^3y^{-10})^4}{4x^3y^{-3}}$

$\frac{2^4x^{12}y^{-40}}{4x^3y^{-3}}$

$\frac{16}{4}x^{12-3}y^{-40+3}$

$4x^9y^{-37} \rightarrow \frac{4x^9}{y^{37}}$

14) $(6xy^9)(4xyz^0)$

$(6 \cdot 4)(x \cdot x)(y^9 \cdot y)(z^0)$

$24x^2y^{10}$

15) $\frac{3}{2x^{-9}}$

$\frac{3x^9}{2}$

16) If you double 2^{200} what is the result? Explain.

$2 \cdot 2^{200} = 2^{201}$

MULT 2^{200} BY 2, ADD EXPONENTS

For problems 17 and 18, simplify the expression. Write your answer in scientific notation.

17) $5.7 \times 10^{100} - 4 \times 10^{98}$

$5.7 \times 10^{100} - 0.04 \times 10^{100}$

$(5.7 - 0.04) \times 10^{100}$

5.66×10^{100}

18) $(5.7 \times 10^{100})(4 \times 10^{98})$

$(5.7 \cdot 4) \times 10^{100+98}$

22.8×10^{198}

2.28×10^{199}

For problems 19 and 20.
Use the figures to the right.



2 7 12

19) If n = figure number, write an expression to determine the number of dots in the nth figure.

$2 + 5(n-1)$

$2 + 5n - 5$

$5n - 3$

20) How many dots are in the 155th figure?

$5(155) - 3$

$775 - 3$

772 DOTS